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siri_chandana@LAPTOP-MMV  ×  +  ▾
siri_chandana@LAPTOP-MMVUDEP:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

siri_chandana@LAPTOP-MMVUDEP:~$ git --version
git version 2.53.0
siri_chandana@LAPTOP-MMVUDEP:~$ mkdir OS_Lab
siri_chandana@LAPTOP-MMVUDEP:~$ cd OS_Lab
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ mkdir Skill1 Skill2
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ ls
Skill1 Skill2
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ |
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siri_chandana@LAPTOP-MMVUDEP:~$ mkdir OS_Lab
siri_chandana@LAPTOP-MMVUDEP:~$ cd OS_Lab
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siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ ls
Skill1 Skill2
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/siri_chandana/OS_Lab/.git/
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ |
```

```
siri_chandana@LAPTOP-MMV  ×  +  ∨
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    printf("Process program executed successfully\n");
    return 0;
}
```

```
nothing to commit (create/copy files and use "git add" to track)
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill1
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill1$ pwd
/home/siri_chandana/OS_Lab/Skill1
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill1$ nano process.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill1$ nano Makefile
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill1$ make
gcc process.c -o process
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill1$ ./process
Process program executed successfully
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill1$ |
```

```
siri_chandana@LAPTOP-MMV  ×  +  ▾
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

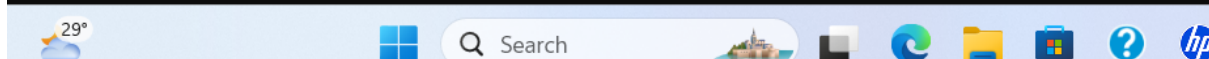
int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("I am the Child Process\n");
    }
    else
    {
        printf("I am the Parent Process\n");
    }

    return 0;
}
```

```
gcc process.c -o process
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ ./process
Process program executed successfully
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ nano fork1.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ gcc fork1.c -o fork1
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ ./fork1
I am the Parent Process
I am the Child Process
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ |
```



```
siri_chandana@LAPTOP-MMV  ×  +  ∨
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else
    {
        printf("Parent Process\n");
        printf("Parent PID = %d\n", getpid());
    }

    return 0;
}
```

```
I am the Child Process
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ nano fork2.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ gcc fork2.c -o fork2
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ ./fork2
Parent Process
Parent PID = 781
Child Process
Child PID = 782
Parent PID = 341
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ |
```

```
siri_chandana@LAPTOP-MMV  X  +  v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Before exec\n");

    execl("/bin/ls", "ls", "-l", NULL);

    printf("This line will not execute if exec is successful\n");

    return 0;
}
```

```
Parent PID = 341
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ nano exec.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ gcc exec.c -o exec
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ ./exec
Before exec
total 84
-rw-r--r-- 1 siri_chandana siri_chandana 71 Aug 28 14:33 Makefile
-rwxr-xr-x 1 siri_chandana siri_chandana 15992 Aug 28 14:38 exec
-rw-r--r-- 1 siri_chandana siri_chandana 207 Aug 28 14:37 exec.c
-rwxr-xr-x 1 siri_chandana siri_chandana 15992 Aug 28 14:35 fork1
-rw-r--r-- 1 siri_chandana siri_chandana 245 Aug 28 14:34 fork1.c
-rwxr-xr-x 1 siri_chandana siri_chandana 16120 Aug 28 14:36 fork2
-rw-r--r-- 1 siri_chandana siri_chandana 358 Aug 28 14:36 fork2.c
-rwxr-xr-x 1 siri_chandana siri_chandana 15952 Aug 28 14:33 process
-rw-r--r-- 1 siri_chandana siri_chandana 104 Aug 28 14:32 process.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ |
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-rw-r--r-- 1 siri_chandana siri_chandana 358 Aug 28 14:36 fork2.c
-rwxr-xr-x 1 siri_chandana siri_chandana 15952 Aug 28 14:33 process
-rw-r--r-- 1 siri_chandana siri_chandana 104 Aug 28 14:32 process.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ pstree -p
systemd(1)─agetty(208)
          └─chronyd-starter(110)─chronyd(224)─chronyd(239)
          └─cron(111)
          └─dbus-daemon(112)
          └─init-systemd(Ub(2))─SessionLeader(340)─Relay(342)(341)─bash(342)─pstree(837)
                                └─init(6)─{init}(7)
                                └─login(343)─bash(418)
                                └─{init-systemd(Ub)}(8)
          └─networkd-dispat(116)
          └─rsyslogd(195)─{rsyslogd}(236)
                        └─{rsyslogd}(237)
                        └─{rsyslogd}(238)
          └─systemd(388)─(sd-pam)(390)
          └─systemd-journal(46)
          └─systemd-logind(151)
          └─systemd-resolve(81)
          └─systemd-udevd(88)
          └─unattended-upgr(261)─{unattended-upgr}(303)
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ |
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    |systemd-resolve(81)
    |systemd-udevd(88)
    |unattended-upgr(261)---{unattended-upgr}(303)
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ strace -e trace=process ./fork1
Command 'strace' not found, but can be installed with:
sudo apt install strace
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ |

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Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ strace -e trace=process ./fork1
execve("./fork1", ["/fork1"], 0x7ffe1c3ee4f0 /* 26 vars */) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0x7e206c9a6a10) = 1026
I am the Child Process
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=1026, si_uid=1000, si_status=0, si_etime=0, si_stime=0} ---
I am the Parent Process
exit_group(0)                                = ?
+++ exited with 0 +++
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill1$ |

```

2.

```

siri_chandana@LAPTOP-MMV  X  +  v
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main()
{
    char command[100];

    while (1)
    {
        printf("MY_SHELL> ");

        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
        {
            break;
        }

        printf("The command entered: %s\n", command);
    }

    return 0;
}

```

```
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill1$ cd ..
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill2
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill2$ pwd
/home/siri_chandana/OS_Lab/Skill2
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill2$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill2$ gcc shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill2$ ./shell
MY_SHELL> hello
The command entered: hello
MY_SHELL> ls
The command entered: ls
MY_SHELL> pwd
The command entered: pwd
MY_SHELL> |
```

```
+++ exited with 0 +++
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill1$ cd ..
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill2
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill2$ pwd
/home/siri_chandana/OS_Lab/Skill2
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill2$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill2$ gcc shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill2$ ./shell
MY_SHELL> hello
The command entered: hello
MY_SHELL> ls
The command entered: ls
MY_SHELL> pwd
The command entered: pwd
MY_SHELL> exit
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill2$ ./shell
MY_SHELL> this is my custom shell
The command entered: this is my custom shell
MY_SHELL> |
```